Multi-UAV Trajectory Planning in Urban Environments with Localization and Communication Fingerprints

Hongbo Zhao, *Senior Member, IEEE*, Zelong Wu, Chen Zhuang, and Zeqi Yin

***Abstract*—In urban low-altitude environments, sudden localization failures and communication interruptions can be induced by building blockages, posing potential risks to the service continuity of uncrewed aerial vehicles (UAVs) in Integrated Localization and Communication (ILAC) systems. To address these challenges, this letter proposes a localization and communication fingerprint (LCF)-aided cooperative trajectory planning framework for multi-UAV systems. An offline LCF map is designed to provide look-ahead service-quality information over a discretized 3D airspace. Then, the planning problem is formulated as a decentralized partially observable Markov decision process (Dec-POMDP) and solved using centralized training and decentralized execution (CTDE) multi-agent reinforcement learning (MARL). Numerical results indicate that the proposed method keeps UAVs away from service-degradation regions and improves localization and communication performance.**

***Index Terms*—Multi-UAV, ILAC, urban low-altitude environment, trajectory planning, MARL.**

I. Introduction

In sixth-generation (6G) wireless systems, integrated localization and communication (ILAC) is emerging as an important paradigm in which localization and communication are supported through shared infrastructure and radio resources [1]. Due to high mobility, flexible deployment, and favorable line-of-sight (LoS) conditions [2]–[4], uncrewed aerial vehicles (UAVs) have been widely investigated as aerial platforms for supporting low-altitude ILAC services.

Complex urban environments impose strict requirements on UAV trajectory planning in ILAC systems. Buildings frequently block air–ground links, generate non-line-of-sight (NLoS) regions, and cause performance degradation in both communication capacity and localization accuracy. In order to resolve such problems, [5] develops a linear-complexity algorithm with fine-grained terrain information, optimizing the 3D urban UAV relay placement under location-dependent LoS conditions. The work in [6] considers UAV minimization and access control for ILAC networks via a greedy search based on criteria importance through intercriteria correlation (CRITIC) and probability-based assignment. These methods provide useful placement guidelines, but they mainly focus on static deployment or assignment decisions rather than online cooperative trajectory planning.

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Benefiting from the development of reinforcement learning (RL), UAVs can learn adaptive decision policies through the interaction with environments. In [7], a transformer-based collaborative RL method is proposed for fluid antenna system (FAS)-enabled 3D UAV positioning with trajectory and antenna-port selection. Reference [8] studies trajectory and resource optimization in mountainous UAV ILAC networks, where geographic occlusion states are aggregated in a hybrid-action soft actor-critic (SAC) framework. [9] investigates multi-agent reinforcement learning (MARL)-based UAV trajectory and beamforming optimization for an integrated sensing and communication (ISAC) system with roaming users.

However, many existing trajectory-planning studies mainly rely on instantaneous UAV observations or reward feedback during online execution, without considering the localization and communication quality of candidate future locations. This may limit the UAVs' ability to anticipate service-degradation regions and lead to suboptimal trajectory decisions. Offline map-aided methods, such as radio-frequency (RF) mapping and geographic-information-aided optimization [10], [11], can provide look-ahead environmental and service-quality information for trajectory planning. Such information reduces repeated per-slot computation and helps UAVs avoid potential localization and communication degradation regions.

Inspired by the map-aided planning idea above, this letter develops a localization and communication fingerprint (LCF)-aided cooperative trajectory planning method for urban multi-UAV ILAC systems. An offline LCF map is constructed from Shannon capacity and a time-difference-of-arrival (TDOA)-based Cramer–Rao lower bound (CRLB) to jointly characterize communication and localization quality over the discretized 3D airspace. Each UAV then observes the LCF values of its current voxel and neighboring feasible voxels for service-aware decision making. Then the constrained planning problem is formulated as a decentralized partially observable Markov decision process (Dec-POMDP) and solved under a centralized training and decentralized execution (CTDE)-based MARL framework. Simulation results show that, compared with geometry-only and raw-metric baselines, the proposed policy further improves communication and positioning performance while preventing UAVs from entering dead zones.

II. System Model and Problem Formulation

A. Urban Environment Model

We consider an urban low-altitude environment consisting of buildings, ground base stations (BSs), and UAVs. The

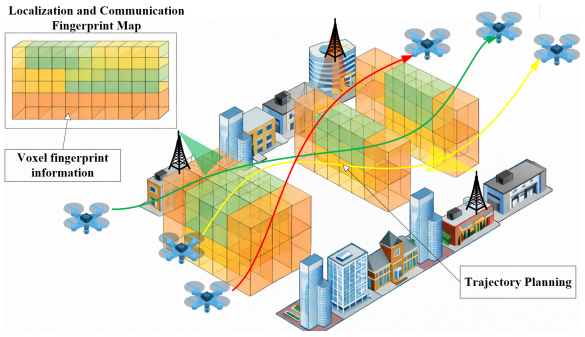


Fig. 1. Urban LCF-aided multi-UAV trajectory planning framework.

building set is $\mathbf{B} = \{b_1, \ldots, b_{|\mathbf{B}|}\}$, where b_j is modeled as an axis-aligned cuboid with center $\mathbf{c}_j^b = (x_j^b, y_j^b, z_j^b)$, dimensions (l_j, w_j, h_j), and region $\Omega(b_j)$. The BS set is $\mathbf{M} = \{m_1, \ldots, m_{|\mathbf{M}|}\}$, where BS m_k is located at $\mathbf{p}_k^m = (x_k^m, y_k^m, z_k^m)$ with transmit power P_k and carrier frequency f_c.

The 3D airspace is discretized into voxels $\mathbf{V} = \{v_1, \ldots, v_{|\mathbf{V}|}\}$. Voxel v_i has center $\mathbf{c}_i^v = (x_i^v, y_i^v, z_i^v)$, edge length Δr, and region $\Omega(v_i)$. The building-occupancy indicator is

$$I_{\text{occ}}(v_i) = \begin{cases} 1, & \exists b_j \in \mathbf{B},\ \Omega(v_i) \cap \Omega(b_j) \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

To account for obstacle avoidance, safety-distance requirements, and low-altitude flight constraints, the feasible voxel set is given by

$$\begin{aligned} \mathbf{V}_{\text{feas}} = \{ & v_i \in \mathbf{V} \mid I_{\text{occ}}(v_i) = 0, \min_{b_j \in \mathbf{B}} d(\Omega(v_i), \Omega(b_j)) \geq d_{\min}^{\text{safety}}, \\ & h_{\min} \leq z_i^v \leq h_{\max} \}, \end{aligned} \tag{2}$$

where $d(\Omega(v_i), \Omega(b_j))$ is the voxel–building distance, $d_{\min}^{\text{safety}}$ is the safety distance, and $[h_{\min}, h_{\max}]$ is the allowable flight-altitude range.

The UAV set is $\mathbf{U} = \{u_1, \ldots, u_{|\mathbf{U}|}\}$. UAV u_n starts from v_n^{start} and targets v_n^{end}. At time slot $t \in \{0, \ldots, T\}$, its position is represented by voxel $v_n(t) \in \mathbf{V}_{\text{feas}}$, and its trajectory is

$$\text{Traj}_n = \{v_n(0), v_n(1), \ldots, v_n(T)\}, \tag{3}$$

where $v_n(0) = v_n^{\text{start}}$, and UAV u_n is required to reach v_n^{end} within the planning horizon. Fig. 1 illustrates the overall framework.

B. LCF Map Construction

To jointly characterize spatial localization and communication performance, we define the LCF of each feasible voxel $v_i \in \mathbf{V}_{\text{feas}}$ as

$$F(v_i) = \{f_{\text{comm}}(v_i), f_{\text{loc}}(v_i)\}, \tag{4}$$

where $f_{\text{comm}}(v_i)$ and $f_{\text{loc}}(v_i)$ denote the communication and localization components of the LCF.

A LoS link between v_i and m_k is assumed available when the corresponding link is free of blockage and lies within the maximum communication range. Let $\mathbf{M}_i$ denote the set of LoS-accessible BSs for voxel v_i.

1) Communication Performance: For the link between v_i and m_k, the signal-to-noise ratio (SNR) is

$$\text{SNR}_{i,k} = \frac{P_k G_{i,k}}{\sigma_0^2}, \tag{5}$$

where σ_0^2 is the noise power, $G_{i,k} = \beta_0\, d_{i,k}^{-\alpha}$ is the channel gain, $d_{i,k} = d(\mathbf{c}_i^v, \mathbf{p}_k^m)$ is the link distance, β_0 is the reference gain, and α is the path-loss exponent. The achievable capacity is $R_{i,k} = B \log_2(1 + \text{SNR}_{i,k})$, where B is the system bandwidth. The communication performance of v_i is then expressed as

$$C(v_i) = \max_{m_k \in \mathbf{M}_i} R_{i,k}. \tag{6}$$

2) TDOA Localization Performance: For positioning reference signal (PRS)-based TDOA ranging, the time-of-arrival (TOA) estimation error variance is $\sigma_{i,k}^2 = c^2 T_{\text{sym}}^2 / (8\pi^2\, \text{SNR}_{i,k} \sum_{j=1}^{N_J} p_j^2 j^2)$, where c is the speed of light, T_{sym} is the symbol duration, and p_j is the power weight of the j-th subcarrier. For each voxel, the visible BSs in $\mathbf{M}_i$ are reindexed and one visible BS is selected as the TDOA reference anchor, denoted by m_1. The CRLB for 3D TDOA localization is [12]

$$\text{CRLB}(v_i) = \text{tr}\big((\mathbf{D}_i^{\text{T}} \mathbf{R}_i^{-1} \mathbf{D}_i)^{-1}\big), \tag{7}$$

where $\mathbf{D}_i$ is the Jacobian matrix whose $(k-1)$-th row is $[(\mathbf{c}_i^v - \mathbf{p}_1^m)/d_{i,1} - (\mathbf{c}_i^v - \mathbf{p}_k^m)/d_{i,k}]^{\text{T}}$ for $k = 2, \ldots, |\mathbf{M}_i|$, and $\mathbf{R}_i$ is the covariance matrix with $[\mathbf{R}_i]_{k,k'} = \sigma_{i,1}^2 + \sigma_{i,k+1}^2 \delta_{k,k'}$, where $\delta_{k,k'}$ is the Kronecker delta. The localization error metric is defined as $L(v_i) = \sqrt{\text{CRLB}(v_i)}$. In addition to requiring $|\mathbf{M}_i| \geq 4$, the Fisher information matrix $\mathbf{D}_i^{\text{T}} \mathbf{R}_i^{-1} \mathbf{D}_i$ must be numerically valid.

3) Normalized Fingerprints: To place the two heterogeneous service metrics on a common scale, normalized fingerprints are constructed through sigmoid mapping. Let $S(x) = 1/(1 + e^{-x})$ and define

$$\Phi(x; \tau, \alpha) = \frac{S(\alpha(x - \tau)) - S(-\alpha\tau)}{1 - S(-\alpha\tau)}. \tag{8}$$

The communication and localization components of the LCF are then given by

$$f_{\text{comm}}(v_i) = \begin{cases} \Phi(C(v_i); C_{\text{th}} - \delta_C, \alpha_C), & |\mathbf{M}_i| \geq 1, \\ 0, & \text{otherwise,} \end{cases} \tag{9}$$

$$f_{\text{loc}}(v_i) = \begin{cases} \Phi(1/L(v_i); 1/(L_{\text{th}} + \delta_L), \alpha_L), & |\mathbf{M}_i| \geq 4, \\ & L(v_i) < \infty, \\ 0, & \text{otherwise,} \end{cases} \tag{10}$$

where C_{th} and L_{th} are the communication-capacity and localization-error thresholds, and $\delta_C, \alpha_C, \delta_L, \alpha_L$ are the offset and slope parameters. Since localization is mapped through inverse error, $1/(L_{\text{th}} + \delta_L)$ is the corresponding $1/L$-domain threshold. When $|\mathbf{M}_i| < 4$ or the CRLB computation is numerically invalid, 3D TDOA localization is regarded as unavailable, and thus $f_{\text{loc}}(v_i) = 0$.

C. UAV Mobility Model

In the discretized airspace, each UAV selects its next position from the adjacent feasible voxel set $\mathbf{V}_n^{\text{adj}}(t)$ of its

current voxel $v_n(t)$. To satisfy speed constraints, the action set includes hovering and 26-neighborhood movements. For voxel edge length Δr and slot duration Δt, the UAV speed is quantized as $V_n(t) \in \{0, \Delta r/\Delta t, \sqrt{2}\Delta r/\Delta t, \sqrt{3}\Delta r/\Delta t\}$.

For energy consumption, the propulsion power $P(V_n(t))$ follows the rotary-wing UAV power model in [13]. The single-step energy is $E_n^{\mathrm{fly}}(t) = P(V_n(t)) \cdot \Delta t$. Since $V_n(t)$ only takes finitely many values, the single-step energy of each movement action can be precomputed offline.

D. Multi-UAV Planning Problem

The multi-UAV cooperative trajectory planning problem aims to determine, for each UAV and each time slot, a feasible next voxel that jointly improves communication performance, localization performance, and energy efficiency while satisfying safety and terminal constraints. The optimization problem is formulated as

$$\begin{aligned}
\max_{\{v_n(t)\}} \quad & \sum_{n=1}^{|\mathbf{U}|} \sum_{t=1}^{T} \Big[w_C f_{\mathrm{comm}}(v_n(t)) + w_L f_{\mathrm{loc}}(v_n(t)) - w_E E_n^{\mathrm{fly}}(t) \Big] \\
\text{s.t.} \quad & v_n(t) \in \mathbf{V}_{\mathrm{feas}} \cap \mathbf{V}_n^{\mathrm{adj}}(t-1), \quad \forall n, t; \\
& v_n(t) \neq v_{n'}(t), \quad \forall n \neq n', t; \\
& \{v_n(t-1), v_n(t)\} \neq \{v_{n'}(t), v_{n'}(t-1)\}, \quad \forall n \neq n', t; \\
& v_n(0) = v_n^{\mathrm{start}}, \quad \exists\, t_n \leq T : \; v_n(t_n) = v_n^{\mathrm{end}}, \quad \forall n; \\
& v_n(t) = v_n^{\mathrm{end}}, \quad \forall t \geq t_n.
\end{aligned} \tag{11}$$

Here, w_E denotes the energy penalty factor, while w_C and w_L denote the communication and localization weight factors, where $w_L = 1 - w_C$. Moreover, $E_n^{\mathrm{fly}}(t)$ is the single-step flight energy. The constraints define feasible adjacent movements, co-voxel collision avoidance, edge-swap collision avoidance, and terminal behavior.

Although the LCF map is constructed offline, each UAV only has access to local information during execution. The problem is therefore a partially observable multi-agent sequential decision-making problem.

III. Proposed Algorithm

Under the CTDE framework, the optimization problem formulated in Section II is solved using multi-agent proximal policy optimization (MAPPO) as the Dec-POMDP solver. The overall architecture is shown in Fig. 2.

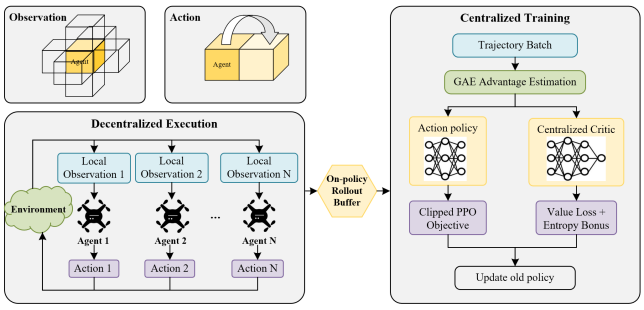


Fig. 2. MAPPO framework for LCF-aided multi-UAV trajectory planning.

1) State Space: At time slot t, the global system state consists of the voxel positions of all UAVs and their LCF values, which is defined as

$$s(t) = \{v_n(t), F(v_n(t))\}_{n=1,\cdots,|\mathbf{U}|}. \tag{12}$$

During decentralized execution, the local observation of UAV u_n is defined as

$$o_n(t) = \{v_n(t), F(v_n(t)), \mathbf{V}_n^{\mathrm{adj}}(t), F(\mathbf{V}_n^{\mathrm{adj}}(t))\}, \tag{13}$$

where $\mathbf{V}_n^{\mathrm{adj}}(t)$ and $F(\mathbf{V}_n^{\mathrm{adj}}(t))$ denote the adjacent feasible voxels of $v_n(t)$ and the corresponding fingerprint information. For neural-network input, the observation uses fixed 27 Moore-neighborhood slots, including hovering; invalid slots are zero-padded and flagged as infeasible.

2) Action Space: Each UAV moves in the discrete grid space, and its action is defined as

$$a_n(t) \in \mathcal{A}_n = \{\mathbf{V}_n^{\mathrm{adj}}(t)\}. \tag{14}$$

The action determines the next voxel from the feasible adjacent voxel set, i.e., $v_n(t) \xrightarrow{a_n(t)} v_n(t+1)$. Although $\mathbf{V}_n^{\mathrm{adj}}(t)$ varies with the current voxel, policy outputs use the same 27 motion slots, and an action mask removes infeasible slots.

3) Reward Design: A team-averaged immediate reward is designed based on voxel fingerprint information. Since the communication and localization components of the LCF are already normalized, they are directly used in the reward, while the energy term is separately normalized. The immediate reward is defined as

$$\begin{aligned}
r(t) = & \frac{1}{|\mathbf{U}|} \sum_{n} \Big\{ X_n(t) \Big[w_C (f_{\mathrm{comm}}(v_n(t)) - 1) \\
& + w_L (f_{\mathrm{loc}}(v_n(t)) - 1) - w_E \widetilde{E}_n(t) - \delta_n^{\mathrm{coll}}(t) \Big] \\
& + r_{\mathrm{arr}} \big[X_n(t-1) - X_n(t) \big] \Big\}.
\end{aligned} \tag{15}$$

Here, $r(t)$ is evaluated after the transition from $v_n(t-1)$ to the landing voxel $v_n(t)$. The first two lines define the per-step service–energy–safety cost gated by $X_n(t)$, and the last term gives a one-time bonus $r_{\mathrm{arr}} > 0$ when X_n switches from 1 to 0. Here, $X_n(t)$ is the arrival-status indicator defined as

$$X_n(t) = \begin{cases} 1, & v_n(t) \neq v_n^{\mathrm{end}}; \\ 0, & \text{otherwise}, \end{cases} \tag{16}$$

which indicates whether UAV u_n remains in an unfinished mission state. When $X_n(t) = 0$, the UAV is regarded as having reached its target voxel; its action is fixed to hovering, and it no longer accumulates communication fingerprint, localization fingerprint, or flight-energy cost.

The normalized energy penalty is defined as $\tilde{E}_n(t) = E_n(t)/E_{\max}$, where $E_n(t)$ is the single-step flight energy and $E_{\max}$ is the maximum single-step energy.

The term $\delta_n^{\mathrm{coll}}(t)$ penalizes unsafe multi-UAV interactions and is defined as

$$\delta_n^{\mathrm{coll}}(t) = \lambda_{\mathrm{coll}} I_{\mathrm{coll}}(v_n(t)), \tag{17}$$

where λ_{coll} is the collision penalty coefficient and $I_{\mathrm{coll}}(v_n(t))$

Algorithm 1 LCF-Aided Multi-UAV Trajectory Planning via MAPPO

Input: LCF map $F(v_i)$, $\mathbf{V}_{\text{feas}}$, start/goal voxels, and MAPPO hyperparameters.
Output: Shared policy network π_θ.
Initialize shared actor θ, centralized critic ϕ, and buffer $\mathcal{D}$.
repeat
Reset $v_n(0) \leftarrow v_n^{\text{start}}$ for all UAVs and clear $\mathcal{D}$.
for $t = 1, \ldots, T$ **do**
Build local observations $o_n(t-1)$ and sample feasible actions $a_n(t-1) \sim \pi_\theta(\cdot \mid o_n(t-1))$.
Execute joint action, observe $r(t)$ and $s(t)$, and store the transition in $\mathcal{D}$.
if all UAVs reach their target voxels **then break**.
end for
Compute GAE advantages $\hat{A}(t)$ and returns $\hat{R}(t)$; update θ and ϕ.
until episode $= Epi_{\max}$

TABLE I
KEY SIMULATION PARAMETERS

Parameter	Value
Environment size	$600 \times 600 \times 150\,\text{m}^3$
Number of buildings	15
Number of BSs	12
Number of UAVs	3
Altitude range	$h_{\min} = 60\,\text{m}$, $h_{\max} = 120\,\text{m}$
Safety distance	$d_{\min}^{\text{safety}} = 5\,\text{m}$
Voxel size	$\Delta r = 20\,\text{m}$
System bandwidth	$B = 10\,\text{MHz}$
Carrier frequency	$f_c = 2.4\,\text{GHz}$
BS transmit power	$P_k = 30\,\text{dBm}$
Path-loss exponent	$\alpha = 2.4$
Noise power	$\sigma_0^2 = -80\,\text{dBm}$
Service weights	$w_C = 0.5$, $w_L = 1 - w_C = 0.5$
Discount factor	$\gamma = 0.99$
GAE parameter	$\lambda = 0.95$
PPO clip range	$\epsilon = 0.2$
Actor learning rate	3×10^{-3}
Critic learning rate	5×10^{-3}
Hidden layer width	128
Training episodes	4000

is the collision indicator function given by

$$I_{\text{coll}}(v_n(t)) = \begin{cases} 1, & \exists n' \neq n, \text{ s.t. } v_n(t) = v_{n'}(t) \text{ or} \\ & v_n(t) = v_{n'}(t-1) \text{ and } v_n(t-1) = v_{n'}(t); \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

The collision indicator captures both co-voxel occupancy and edge-swap crossing between adjacent time slots. In MAPPO implementation, action masks enforce feasible single-UAV motions, while multi-UAV conflicts are handled by this penalty.

Based on the above designs, the MAPPO algorithm proceeds as follows, where Generalized advantage estimation (GAE) is used to compute policy-gradient advantages.

IV. EXPERIMENTAL EVALUATION

Simulations are conducted in the urban low-altitude scenario of Section II. The communication and localization sigmoid centers are calibrated from the medians of $C(v_i)$ and $1/L(v_i)$ over feasible voxels, so that their transition regions cover about 30%–70% of the feasible space. Key parameters follow representative UAV trajectory-design and ILAC studies [8], [14], as summarized in Table I.

Fig. 3 compares representative 3D trajectories and their communication capacity (Mbit/s) and localization error (m).

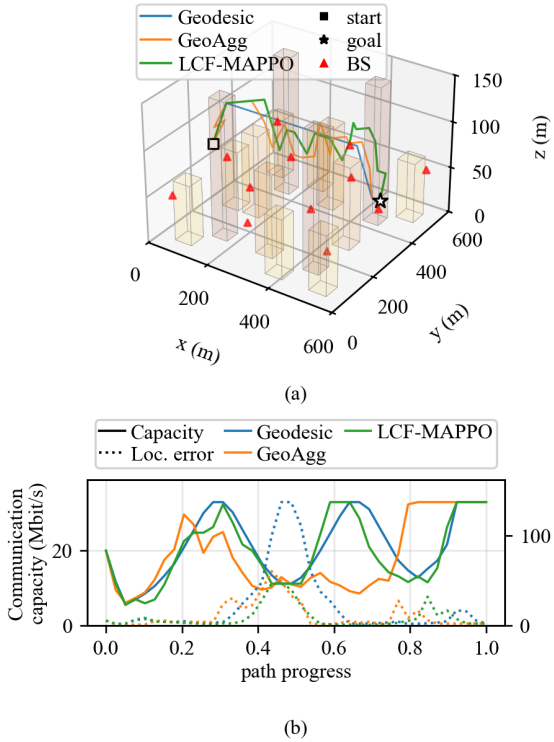


Fig. 3. Three-way method comparison: (a) representative 3D trajectories through the urban environment; (b) communication capacity (solid) and localization error (dotted) along the path.

TABLE II
MEAN SERVICE QUALITY OF COMPARED METHODS

Method	Capacity (Mbit/s)	Error (m)	Path steps
Greedy geodesic	18.69	38.72	23
GeoAgg-style raw-metric RL	20.69	19.34	29
LCF (proposed)	20.74	11.14	24

Table II reports averaged service values and path steps. The geodesic greedy baseline is LCF-blind and follows geometry-only shortest paths. The GeoAgg-style raw-metric RL baseline, adapted from [8], uses a log-potential reward on raw physical quantities rather than the LCF map. All three methods achieve full arrival without collisions in the discrete collision model. The proposed complete LCF-aware design yields the lowest mean localization error while maintaining a mean capacity close to the raw-metric RL baseline. The representative geodesic trajectory crosses the dead zone and shows a larger localization error, whereas the raw-metric RL baseline has no fingerprint-based observation. The observed performance is consistent with the use of adjacent-voxel service-quality information, which may facilitate service-aware trajectory decisions.

Fig. 4(a) compares the MARL algorithms. MAPPO gives the highest mean team return among the tested algorithms. Independent proximal policy optimization (IPPO) removes the centralized critic and gives a lower return, indicating the role of CTDE under partial observability and multi-UAV coupling. Multi-agent advantage actor–critic (MAA2C) reaches a competitive average return but exhibits larger fluctuations. Fig. 4(b) shows that the grid with $\Delta r = 20\,\text{m}$ converges fastest because its compact state–action space reduces sample complexity,

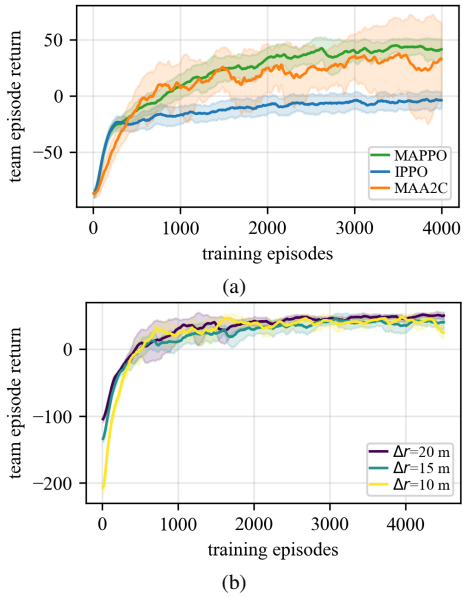


Fig. 4. Training dynamics: (a) algorithmic comparison among MAPPO, MAA2C and IPPO; (b) voxel resolution sweep with $\Delta r = 20/15/10\,\mathrm{m}$. Shaded bands denote mean $\pm$ one standard deviation across seeds.

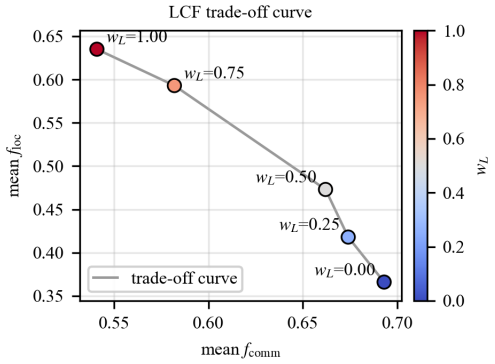


Fig. 5. Localization–communication tradeoff curve obtained by sweeping the localization weight w_L.

whereas finer grids require more exploration. Thus, voxel resolution trades spatial fidelity for learning complexity.

Fig. 5 shows that the localization–communication balance can be adjusted by sweeping w_L. The resulting tradeoff curve is monotonic over the tested range: increasing w_L from 0 to 1 improves f_{loc} from 0.37 to 0.64, whereas f_{comm} decreases from 0.69 to 0.54. All UAVs reach their targets at every operating point.

V. Conclusion

This letter presents an LCF-aided cooperative trajectory planning framework for multi-UAV ILAC systems in urban low-altitude environments. The offline LCF map provides localization and communication information from the current and adjacent voxels, which MAPPO utilizes for cooperative planning. Simulation results demonstrate that the proposed LCF-aware planner outperforms the geometry-only and raw-metric RL baselines by achieving lower localization error at comparable communication capacity. Moreover, sweeping the localization weight reveals an effective localization–communication tradeoff.

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